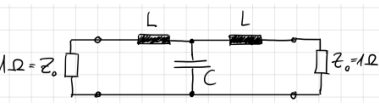


1 Synthe Passiver Schaltungen

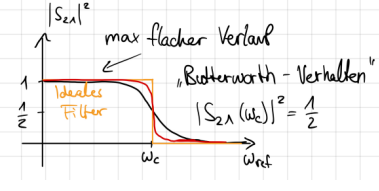
1.1 Prototypfilter/Referenzfilter



$Z_0 = 1\Omega$ Normierung: bei

$\omega_{ref} = \omega_c$:

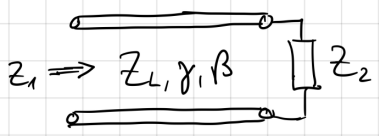
- $\omega_c C = 2\Omega^{-1}$
- $\omega_c L = 1\Omega$



$$|S_{21}(\omega_c)|^2 = \frac{1}{2}$$

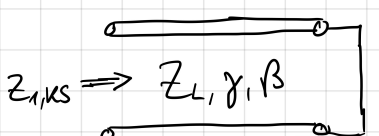
1.2 Leitungsfiter

für HF-Filter Übergang zu Leitungselementen



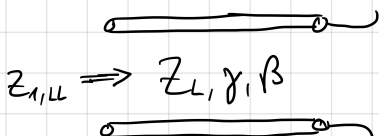
$$Z_1 = Z_L \frac{Z_2 + jZ_L \tan(\beta l)}{Z_L + jZ_2 \tan(\beta l)}$$

$$\beta l = \Theta = \frac{2\pi}{\lambda} l = \frac{2\pi}{\lambda} \frac{\lambda}{4} \frac{\omega}{\omega_0} = \frac{\pi}{2} \frac{\omega}{\omega_0}$$



Kurzschliessen von Z_2

$$Z_2 = 0 \Rightarrow Z_{1,KS} = jZ_L \tan(\beta l) \hat{=} \text{induktiv } Z_L \hat{=} \tilde{L}$$



Leerlauf

$$Z_2 \rightarrow \infty \Rightarrow Z_{1,LL} = Z_L \frac{1}{j \tan(\beta l)} \hat{=} \text{kapazitiv } Z_L \hat{=} \tilde{C}$$

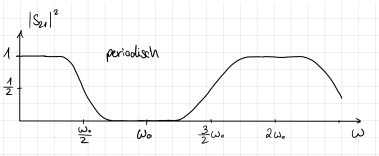
$$\frac{\omega_c}{\omega_{ref} C} \Rightarrow \tilde{C} = C$$

für $L = \tilde{L}$ und $C = \tilde{C}$:

$$\frac{\omega_{ref}}{\omega_c} = \tan(\beta l) = \tan\left(\frac{\pi}{2} \frac{\omega}{\omega_c}\right) \Leftrightarrow \omega = \omega_0 \frac{2}{\pi} \arctan\left(\frac{\omega_{ref}}{\omega_c}\right) \quad (1)$$

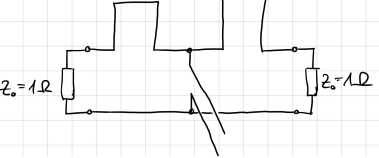
$\omega_{ref} \hat{=}$ Frequenz des Referenzfilter

ω ist die Frequenz des Leitungsfilters



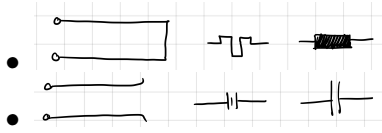
- $\omega_{ref} = 0 \Rightarrow \omega = 0$
- $\omega_{ref} \rightarrow \infty \Rightarrow \omega = \omega_0 \frac{\pi}{2} \frac{2}{\pi} = \omega_0$
- $\omega_{ref} = \omega_c \Rightarrow \omega = \omega_0 \frac{2}{\pi} \arctan(1) = \frac{\omega_0}{2}$

2Draht-ESB:



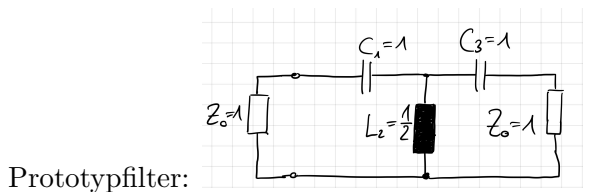
Konzentrierte Elemente ersetzen durch Stichleitungen.

- Induktivitäten: Am Ende KS Stichleitungen

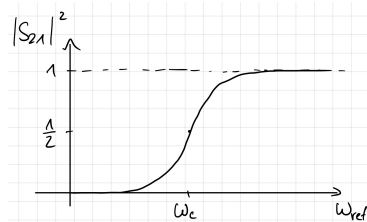


- Kapazitäten: Am Ende LL Stichleitungen

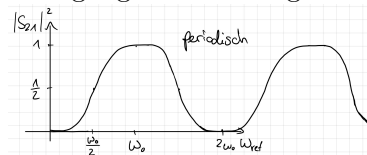
1.3 Butterworth-Hochpassfilter



Prototypfilter:

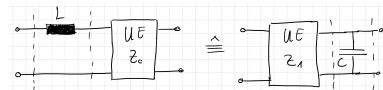


Übergang zum Leitungsfiter



bei Mikrostreifenleitungen sind keine seriel liegenden Stichleitungen möglich.

1.4 Kuroda Identitäten



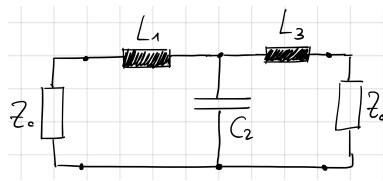
$S = j \tan \beta L$ Richardstransformation

$$\frac{1}{\sqrt{1-s^2}} \begin{bmatrix} 1 & sL \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & sZ_0 \\ \frac{s}{Z_0} & 1 \end{bmatrix} \stackrel{!}{=} \frac{1}{\sqrt{1-s^2}} \begin{bmatrix} 1 & sZ_1 \\ \frac{s}{Z_1} & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ sC & 1 \end{bmatrix}$$

$$\begin{bmatrix} q + s^2 \frac{L}{Z_0} & s(L + Z_0) \\ \frac{s}{Z_0} & 1 \end{bmatrix} \stackrel{!}{=} \begin{bmatrix} 1 + s^2 Z_1 C & sZ_1 \\ s(\frac{1}{Z_1} + C) & 1 \end{bmatrix}$$

- $\frac{L}{Z_0} = Z_1 C$
- $Z_1 = L + Z_0$
- $\frac{1}{Z_0} = \frac{1}{Z_1} + C$
- $n = 1 + \frac{L}{Z_0}$
- $Z_1 = nZ_0$
- $C = \frac{n-1}{n} \frac{1}{Z_0}$

1.4.1 Leitungsfiter



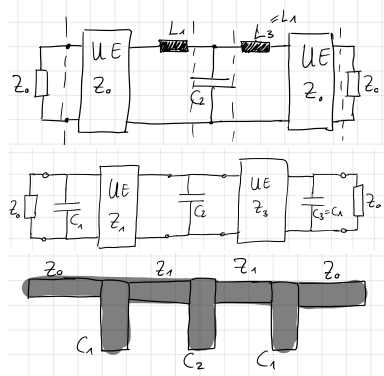
Realisierung in

Mikrostreifenleitungen. Problem:

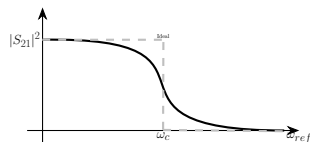
- serielle Elemente nicht realisierbar.
- Leitungsverzweigung ggf. problematisch

Anwendung der Kuroda-Identitäten.

Einfügen von Leitungselementen mit Z_0 von außen. $\Rightarrow$ keine Veränderung des Betrages, nur Phaseveränderung.



1.5 LC- Tiefpass mit Butterworthverhalten



- $|S_{21}|^2$ max flach bei $\omega=0$
- $|S_{21}(\omega_c)|^2 = \frac{1}{2}$

Annahme: Verlustlosigkeit $\Rightarrow |S_{11}|^2 + |S_{21}|^2 = 1$

$$\Rightarrow |S_{21}|^2 = \frac{|S_{21}|^2}{|S_{21}|^2 + |S_{11}|^2} = \frac{1}{1 + \frac{|S_{11}|^2}{|S_{21}|^2}} = \frac{1}{1 + |\Sigma_{21}|^2} = \frac{1}{1 + Q_m(y)}$$

mit $y = \left(\frac{\omega_{ref}}{\omega_c}\right)^2$ und $Q_m = a_0 + a_1 y + a_2 y^2 + \dots + a_m y^m$

$(m-1)$: Ableitung von $|S_{21}|^2$ bei $y=0 \Rightarrow$ verschwinden

- $\left(\frac{1}{1+Q_m(y)}\right)' \big|_{y=0} \stackrel{!}{=} 0 \Rightarrow a_1 = 0$
- $\left(\frac{1}{1+Q_m(y)}\right)'' \big|_{y=0} \stackrel{!}{=} 0 \Rightarrow a_2 = 0$
- $\Rightarrow a_m \neq 0$

$$Q_m(y) = a_0 + a_m y^m$$

$$1. |S_{21}(0)|^2 \stackrel{!}{=} 1 \Rightarrow \frac{1}{1+a_0} = 1 \Rightarrow a_0 = 0$$

$$2. |S_{21}(1)|^2 \stackrel{!}{=} \frac{1}{2} \Rightarrow \frac{1}{1+a_m} = \frac{1}{2} \Rightarrow a_m = 1 \text{ bei } \omega_{ref} = \omega_c$$

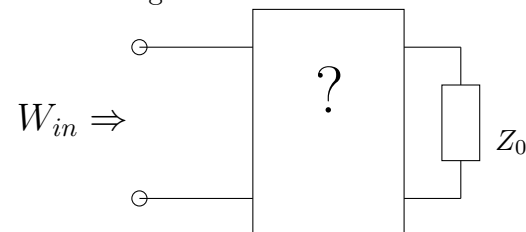
$$|S_{21}|^2 = \frac{1}{1+y^m}$$

mit $y = \left(\frac{\omega_{ref}}{\omega_c}\right)^2 \Rightarrow$ Leitungsfiler $s \hat{=} j\omega_{ref}$, $s_c \hat{=} j\omega_c$

$$|S_{21}|^2 = \frac{1}{1 + \left(\frac{s}{s_c}\right)^{2m}}$$

1.5.1 Erklärung

geg.: beschreibende Funktion (hier $|S_{21}|^2$) $\Rightarrow$ Ziel: Realisierung mit L und C



$$W_{in}(s) = \frac{1-S_{11}(s)}{1+S_{11}(s)}$$

$$|S_{11}|^2 = 1 - |S_{21}|^2 = 1 - \frac{1}{1 + \left(\frac{s}{s_c}\right)^{2m}} = \frac{\left(\frac{s}{s_c}\right)^{2m}}{1 + \left(\frac{s}{s_c}\right)^{2m}}$$

$$|S_{11}|^2 = S_{11}(s)S_{11}(-s)$$

$S_{11}(s)$ Pole in linker s-HE für Stabilität

1.5.2 Beispiel

$$s_c = j, m = 3$$

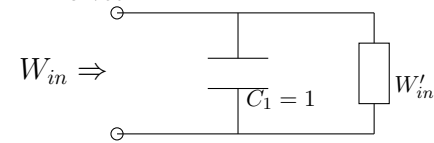
$$|S_{11}|^2 = \frac{(-jS)^6}{1+(-jS)^6} = \frac{-s^6}{1-s^6} = S_{11}(s)S_{11}(-s) =$$

$$\frac{\mp s^3}{-s^3+2s^2-2s+1} \frac{\pm s^3}{s^3+2s^2+2s+1} \quad S_{11}(s) \text{ ist zweiter frac}$$

$$W_{in}(s) = \frac{1+S_{11}(s)}{1-S_{11}(s)} = \frac{1 + \frac{-s^3}{s^3+2s^2+2s+1}}{1 - \frac{-s^3}{s^3+2s^2+2s+1}}$$

$$\text{Wahl des negative VZ} \Rightarrow W_{in}(s) = \frac{s^3+2s^2+2s+1-s^3}{s^3+2s^2+2s+1+s^3}$$

$$\bullet \lim_{s \rightarrow \infty} W_{in}(s) = \frac{1}{s} \Rightarrow C = 1$$



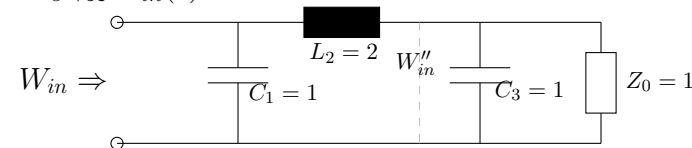
$$W_{in}(0) = 1 \hat{=} \text{Durchlassbereich}$$

$$W_{in} = \frac{1}{V_{in}}$$

$$\text{Wechsel in Admittanzebene: } V_{in} = sC_1 + V'_{in} \Leftrightarrow V'_{in} = V_{in} - s = \frac{2s^3+2s^2+2s+1-2s^3-2s^2-s}{2s^2+2s+1} = \frac{s+1}{2s^2+2s+1}$$

$$W'_{in} = \frac{2s^2+2s+1}{s+1}$$

$$\lim_{s \rightarrow \infty} W'_{in}(s)' = 2s \Rightarrow \text{induktiv } L = 2$$



$$W'_{in} = sL_2 + W''_{in} \Rightarrow W''_{in} = W'_{in} - sL_2 = \frac{2s^2+2s+1-2s^3-2s}{s+1} = \frac{1}{s+1}$$

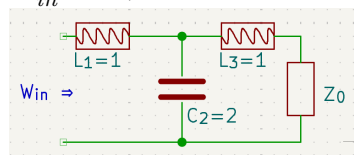
$$V''_{in} = \frac{1}{W''_{in}} = s + 1 \hat{=} \text{Parallelschaltung } C_3 = 1 \text{ und } Z_0 = 1$$

Wahl des positiven VZ:

$$W_{in} = \frac{s^3+2s^2+2s+1+s^3}{s^3+2s^2+2s+1-s^3} = \frac{2s^3+2s^2+2s+1}{2s^2+2s+1} \quad \text{Noch immer TF, da}$$

$$W_{in}(0) = 0$$

- $\lim_{s \rightarrow \infty} W_{in} = s \Rightarrow \text{induktiv } L_1 = 1$
- $W_{in} = sL_1 + W'_{in} \Rightarrow W'_{in} = \frac{2s^3+2s^2+2s+1-2s^3-2s^2-s}{2s^2+2s+1} = \frac{s+1}{2s^2+2s+1}$
- $\lim_{s \rightarrow \infty} W'_{in} = \frac{1}{2s} \Rightarrow \text{kapazitiv } C_2 = 2$
- $V'_{in} = sC_2 + V''_{in} \Rightarrow V''_{in} = \frac{2s^2+2s+1-2s^2-2s}{s+1} = \frac{1}{s+1}$
- $W''_{in} = s + 1$



-

1.6 LC-Hochpassfilter mit Butterworthverhalten

vom TP: $|S_{21}|^2 = \frac{1}{1 + \left(\frac{s}{s_c}\right)^{2m}}$ Übergang zum HP durch

$$\text{substitution } s \rightarrow \frac{1}{s} \quad \text{HP: } |S_{21}|^2 = \frac{1}{1 + \left(\frac{s_c}{s}\right)^{2m}}$$

1.6.1 Beispiel

$$m = 3, s_c = j$$

$$|S_{21}|^2 = \frac{1}{1 + \frac{-1}{s^6}} = \frac{s^6}{s^6-1}$$

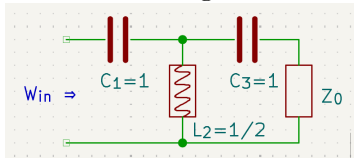
$$|S_{11}|^2 = 1 - |S_{21}|^2 = \frac{-1}{s^6-1} = \frac{1}{1-s^6} = S_{11}(s)S_{11}(-s)$$

$$S_{11}(s) = \frac{\pm 1}{s^3+2s^2+2s+1}$$

Wahl des pos. VZ

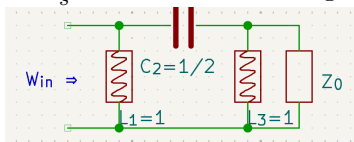
- $W_{in} = \frac{1+S_{11}}{1-S_{11}} = \frac{s^3+2s^2+2s+1+1}{s^3+2s^2+2s+1-1} = \frac{s^3+2s^2+2s+2}{s^3+2s^2+2s}$
- HP da $\lim_{s \rightarrow \infty} = 1$
- $\lim_{s \rightarrow 0} W_{in} = \frac{2}{2s} = \frac{1}{s} \Rightarrow C = 1$
- Kapazität is seriel
- $W'_{in} = W_{in} - \frac{1}{s} = \frac{s^3+2s^2+2s+2-s^2-2s-2}{s^3+2s^2+2s} = \frac{s^3+s^2}{s^3+2s^2+2s} = \frac{s^2+s}{s^2+2s+2}$

- $\lim_{s \rightarrow 0} W'_{in} = \frac{s}{2} \Rightarrow L = \frac{1}{2}$
- Induktivität ist parallel
- $V''_{in} = V'_{in} - sL = \frac{s^2+2s+2-2s-2}{s^2+s} = \frac{1}{1+s} \Rightarrow$
Reihenschaltung aus $C = 1$ und $Z_0 = 1$

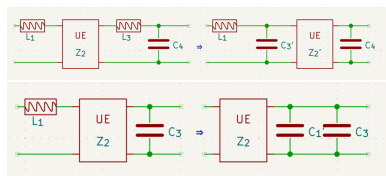


Wahl des neg. VZ

- $W_{in} = \frac{1+S_{11}}{1-S_{11}} = \frac{s^3+2s^2+2s+1-1}{s^3+2s^2+2s+1+1} = \frac{s^3+2s^2+2s}{s^3+2s^2+2s+2}$
- HP da $\lim_{s \rightarrow \infty} = 1$
- $\lim_{s \rightarrow 0} W_{in} = \frac{2s}{2} = \frac{s}{1} \Rightarrow L = 1$
- Induktivität ist parallel
- $V'_{in} = V_{in} - \frac{1}{s} = \frac{s^3+2s^2+2s+2-s^2-2s-2}{s^3+2s^2+2s} = \frac{s^3+s^2}{s^3+2s^2+2s} = \frac{s^2+s}{s^2+2s+2}$
- $\lim_{s \rightarrow 0} W'_{in} = \frac{2}{s} \Rightarrow C = \frac{1}{2}$
- Kapazität ist seriel
- $W''_{in} = W'_{in} - \frac{2}{s} = \frac{s^2+2s+2-2s-2}{s^2+s} = \frac{s^2}{s^2+s} = \frac{s}{s+1} \Rightarrow V''_{in} = 1 + \frac{1}{s} \Rightarrow$ Parallelschaltung aus $L_3 = 1$ und $Z_0 = 1$



1.7 Filter aus Induktivitäten und Kapazitäten sowie Leitungselementen



$\Rightarrow$ nicht redundant

$\Rightarrow$ redundant

Beschriebene Funktion für Butterworth

$$[K_{UE}] = \frac{1}{\sqrt{1-s^2}} \begin{bmatrix} 1 & sZ_2 \\ \frac{s}{Z_2} & 1 \end{bmatrix}$$

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} 1 & sL_1 \\ 0 & 1 \end{bmatrix} \frac{1}{\sqrt{1-s^2}} \begin{bmatrix} 1 & sZ_2 \\ \frac{s}{Z_2} & 1 \end{bmatrix} \begin{bmatrix} 1 & sL_3 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ sC_4 & 1 \end{bmatrix}$$

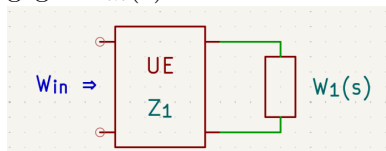
$$\left(\frac{1}{\sqrt{1-s^2}} \right)^n P_{m+n}(s)$$

n: Anzahl Leitungselementen m: Anzahl Induktivitäten und Kapazitäten

$$|S_{21}(s)|^2 = \frac{1}{1+|\Sigma_{21}(s)|^2}$$

$$|\Sigma_{21}(s)|^2 = \left(\frac{s^2}{s_c^2} \right)^m \left(\frac{s^2}{s_c^2} \right)^n \frac{(1-s_c^2)^n}{(1-s^2)^n} \text{ TP mit Butterworth}$$

geg:: $W_{in}(s)$:



$$W_{in} = Z_1 \frac{W_1(s)+sZ_1}{Z_1+sW_1(s)}$$

$$\text{Auflösen nach } W_1(s): W_1(s) = \frac{Z_1(sZ_1-W_{in})}{sW_{in}-Z_1}$$

$$W_{in}(s=1) = Z_1 \frac{W_1(1)+Z_1}{Z_1+W_1(1)} = Z_1$$

Zum Zeitpunkt $t = 0s \Rightarrow$ Spannungssprung am Leitungsanfang $\rightarrow t = 0 \rightarrow$ Verhältnis von Spannung zu Strom $\hat{=}$ Leitungswellenwiderstand Z_1

Grenzwertsatz der Laplace-Transformation:

$$f(0+) = \lim_{p \rightarrow \infty} pF(p) \text{ mit } p = j\omega \Leftrightarrow \omega = -jp$$

$$\text{mit } s = j \tan\left(\frac{\pi}{2} \frac{\omega}{\omega_0}\right) = j \tan\left(\frac{\pi}{2} \left(\frac{-jp}{\omega_0}\right)\right)$$

$$\lim_{p \rightarrow \infty} j \tan\left(\frac{\pi}{2} \left(\frac{-jp}{\omega_0}\right)\right) = 1 \Rightarrow p \rightarrow \infty \hat{=} s \rightarrow 1$$

$$W'_{in}(s) = W_{in}(1) \frac{sW_{in}(1)-W_{in}(s)}{sW_{in}(s)-W_{in}(1)}$$

Zähler und Nenner NS bei $s = 1$ und $s = -1s$

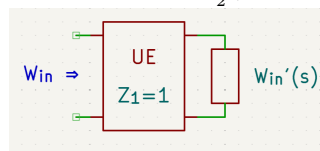
$$(1+s)(1-s) = (1-s^2)$$

1.7.1 Beispiel

$$W_{in}(s) = \frac{1+3s+\frac{1}{2}s^2}{1+\frac{3}{2}s+2s^2}$$

1. Ist ein UE enthalten? $W_{in}(1) \stackrel{!}{=} -W_{in}(-1)$

- $W_{in}(1) = \frac{1+3+\frac{1}{2}}{1+\frac{3}{2}+2} = 1 = Z_1$
- $-W_{in}(-1) = -\frac{1-3+\frac{1}{2}}{1-\frac{3}{2}+2} = 1$

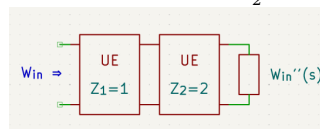


2. Abspaltung

$$W'_{in}(s) = \frac{s - \frac{1+3s+\frac{1}{2}s^2}{1+\frac{3}{2}s+2s^2}}{\frac{1+3s+\frac{1}{2}s^2}{1+\frac{3}{2}s+2s^2} - 1} = \frac{s + \frac{3}{2}s^2 + 2s^3 - 1 - 3s - \frac{1}{2}s^2}{s + 3s^2 + \frac{1}{2}s^3 - 1 - 3s - 2s^2} = \frac{2s^3 + s^2 - 2s - 1}{\frac{1}{2}s^3 + s^2 - \frac{1}{2}s - 1} = \frac{2s+1}{\frac{1}{2}s+1}$$

3. Ist UE enthalten

- $W'_{in}(1) = \frac{3}{\frac{1}{2}} = 2 = Z_2$
- $-W_{in}(-1) = -\frac{-1}{\frac{1}{2}} = 2$



4. Abspaltung

$$W''_{in}(s) = 2 \frac{s^2 - \frac{2s+1}{\frac{1}{2}s+1}}{s \frac{2s+1}{\frac{1}{2}s+1} - 2} = 2 \frac{s^2 + 2s - 2s - 1}{2s^2 + s - s - 2} = \frac{s^2 - 1}{s^2 - 1} = 1$$

1.7.2 Beispiel

$$W_{in}(s) = \frac{1+3s+\frac{1}{2}s^2}{1+\frac{1}{2}s}$$

1. $W_{in}(1) = -W_{in}(-1)$?:

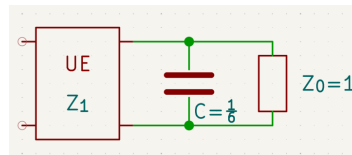
- $W_{in}(1) = \frac{1+3+\frac{1}{2}}{1+\frac{1}{2}} = 3 = Z_1$
- $W_{in}(-1) = \frac{1-3+\frac{1}{2}}{1-\frac{1}{2}} = -3$

2. Abspaltung:

$$W'_{in}(s) = \frac{3(3s - \frac{1+3s+\frac{1}{2}s^2}{1+\frac{1}{2}s})}{s \frac{1+3s+\frac{1}{2}s^2}{1+\frac{1}{2}s} - 3} = 3 \frac{3s + \frac{3}{2}s^2 - 1 - 3s - \frac{1}{2}s^2}{s + 3s^2 + \frac{1}{2}s^3 - 3 - \frac{3}{2}s} = 3 \frac{s^2 - 1}{\frac{1}{2}s^3 + 3s^2 - \frac{1}{2}s - 3} = \frac{3}{2} \frac{s^2 - 1}{s^3 + 6s^2 - s - 6} = \frac{3}{2} \frac{s^2 - 1}{(s^2 - 1)(s + 6)} = \frac{3}{2} \frac{1}{s + 6}$$

3. $V_1(s)' = \frac{1}{6}s + 1$: Parallelschaltung aus C und Z_0

4.



5.

2. Weg: Erst Induktivität/Kapazität abspalten.

1. TP: $\lim_{s \rightarrow 0} = 0$
2. $\lim_{s \rightarrow \infty} W_{in} = s \Rightarrow L_1 = 1$ Induktivität

$$3. W'_{in} = W_{in} - s = \frac{1+3s+\frac{1}{2}s^2-s-\frac{1}{2}s^2}{1+\frac{1}{2}s} = \frac{1+2s}{1+\frac{1}{2}s} = \frac{2+4s}{2+s}$$

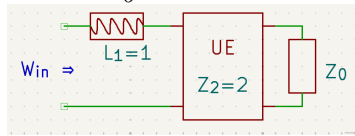
$$4. W'_{in}(1) = -W'_{in}(-1)?:$$

- $W'_{in}(1) = \frac{2+4}{2+1} = 2 = Z_2$
- $W'_{in}(-1) = \frac{2-4}{2-1} = -2$

5. Abspaltung:

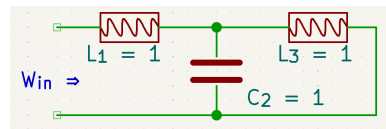
$$W''_{in}(s) = \frac{2s - \frac{2+4s}{2+s}}{s(\frac{2+4s}{2+s}) - 2} = \frac{4s+2s^2-2-4s}{2s+4s^2-4-2s} = \frac{2s^2-2}{4s^2-4} = 1$$

6. $V_1(s)' = \frac{1}{6}s + 1$: Parallelschaltung aus C und Z_0



7.

1.7.3 Beispiel



$$W_{in} = s + \frac{\frac{1}{s}}{\frac{1}{s} + s} = s + \frac{s}{s^2+1} = \frac{s(s^2+2)}{s^2+1} = \frac{s^3+2s}{s^2+1}$$

Überprüfung mit Realisierung

1. $W_{in}(1) = -W_{in}(-1)?:$

- $W_{in}(1) = \frac{1(1+2)}{1+1} = \frac{3}{2} = Z_1$
- $W_{in}(-1) = \frac{-1+2}{-1+1} = -\frac{3}{2}$

2. Abspaltung:

$$W'_{in}(s) = \frac{3}{2} \frac{s^{\frac{3}{2}} - \frac{s(s^2+2)}{s^2+1}}{s(\frac{s(s^2+2)}{s^2+1}) - \frac{3}{2}} = \frac{3}{2} \frac{\frac{3}{2}s^3 + \frac{3}{2}s - s^3 - 2s}{s^4 + 2s^2 - \frac{3}{2}s^2 - \frac{3}{2}} =$$

$$\frac{3}{2} \frac{\frac{1}{2}s^3 - \frac{1}{2}s}{s^4 + \frac{1}{2}s^2 - \frac{3}{2}} = \frac{3}{2} \frac{s^3 - s}{2s^4 + s^2 - 3} = \frac{3}{2} \frac{s^2-1}{s^2+3} \frac{s}{2s^2+3} = \frac{3s}{4s^2+6}$$

3. $W'_{in}(1) = -W'_{in}(-1)?:$

- $W'_{in}(1) = \frac{3}{10} = Z_2$
- $W'_{in}(-1) = \frac{-3}{10}$

4. Abspaltung: $W''_{in}(s) = \frac{3}{10} \frac{\frac{3}{10}s - \frac{3s}{4s^2+6}}{s(\frac{3s}{4s^2+6}) - \frac{3}{10}} = \frac{3}{10} \frac{\frac{6}{5}s^3 + \frac{9}{5}s - 3s}{3s^2 + \frac{6}{5}s^2 - \frac{9}{5}} =$

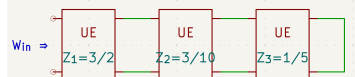
$$\frac{3}{10} \frac{\frac{6}{5}s^3 - \frac{6}{5}s}{\frac{9}{5}s^2 - \frac{9}{5}} = \frac{1}{5} \frac{s(s^2-1)}{s^2-1} = \frac{1}{5}s$$

5. $W'_{in}(1) = -W'_{in}(-1)?:$

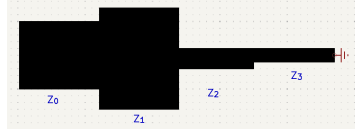
- $W'_{in}(1) = \frac{1}{5} = Z_3$
- $W'_{in}(-1) = \frac{-1}{5}$

6. Abspaltung: $W''_{in}(s) = \frac{1}{5} \frac{\frac{1}{5}s - \frac{1}{5}s}{s(\frac{1}{5}s) - \frac{1}{5}} = \frac{1}{5} \frac{0}{\frac{1}{5}s^2 - \frac{1}{5}} = 0 \Rightarrow$

Kurzschluss

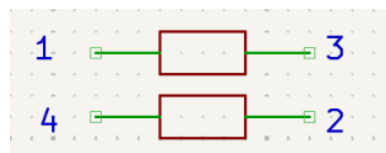


7.



8.

1.8 Gekoppelte Streifenleitungen



geg. Z_e, Z_d

$$[U] = [Z][I]$$

$$\bullet Z_{11} = Z_{22} = Z_{33} = Z_{44} = \frac{Z_e + Z_d}{2s}$$

- $Z_{12} = Z_{21} = Z_{34} = Z_{43} = \frac{Z_e + Z_d}{2s} \sqrt{1 - s^2}$
- $Z_{13} = Z_{31} = Z_{24} = Z_{42} = \frac{Z_e + Z_d}{2s} \sqrt{1 - s^2}$
- $Z_{14} = Z_{41} = Z_{23} = Z_{32} = \frac{Z_e + Z_d}{2s}$



4 $\rightarrow$ KS $\Rightarrow U_4 = 0$ und 3 $\rightarrow$ LL $\Rightarrow I = 0$

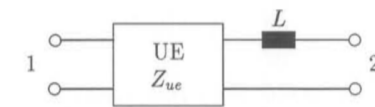
$$\begin{bmatrix} U_1 \\ U_2 \\ U_3 \\ 0 \end{bmatrix} = [Z] \begin{bmatrix} I_1 \\ I_2 \\ 0 \\ I_4 \end{bmatrix} \quad 4. \text{ Gl.}:$$

$$0 = Z_{41}I_1 + Z_{42}I_2 + Z_{44}I_4 \Leftrightarrow I_4 = \frac{Z_{41}}{Z_{44}}I_1 - \frac{Z_{42}}{Z_{44}}I_2$$

I_4 in 1 und 2 Gl. einsetzen:

$$\begin{bmatrix} U_1 \\ U_2 \end{bmatrix} = \begin{bmatrix} Z_{11} - \frac{Z_{14}Z_{41}}{Z_{44}} & Z_{12} - \frac{Z_{14}Z_{42}}{Z_{44}} \\ Z_{21} - \frac{Z_{24}Z_{41}}{Z_{44}} & Z_{22} - \frac{Z_{24}Z_{42}}{Z_{44}} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

$$[U] = [Z'] [I]$$



$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \frac{1}{\sqrt{1-s^2}} \begin{bmatrix} 1 & sZ_{ue} \\ \frac{2}{Z_{ue}} & 1 \end{bmatrix} \begin{bmatrix} 1 & sL \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & s(L + Z_{ue}) \\ \frac{s}{Z_{ue}} & \frac{s^2L}{Z_{ue}} + 1 \end{bmatrix}$$

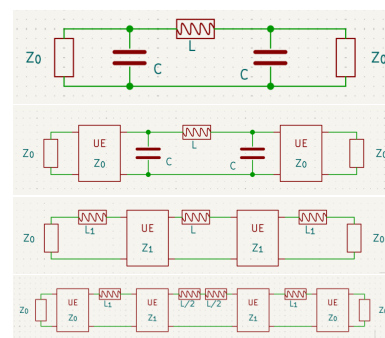
für Koeffizientenvergleich: $[Z'']$

$$[Z''] = \begin{bmatrix} \frac{A}{C} & \frac{AD-BC}{C} \\ \frac{1}{C} & \frac{D}{C} \end{bmatrix} \Rightarrow Z''_{11} = \frac{A}{C} \stackrel{!}{=} \frac{Z_{ue}}{s}$$

$$Z'_{11} = \frac{2Z_eZ_d}{s(Z_e+Z_d)} \stackrel{!}{=} Z''_{11} \Rightarrow Z_{ue} = \frac{2Z_eZ_d}{Z_e+Z_d}$$

$$L = \frac{(Z_e - Z_d)^2}{2(Z_e + Z_d)}$$

1.8.1 Beispiel

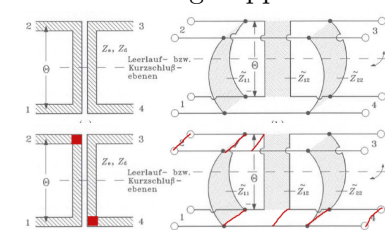


Die Drauf sicht der Realisierung:

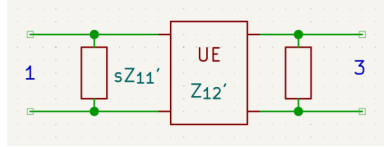


1.9 Beispiel

ESB mit 3 ungekoppelten Leitungselementen

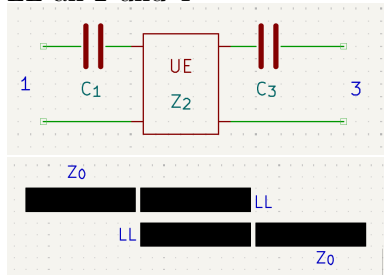


KS and 2 und 4



Fehlt Bild

LL an 2 und 4



2. ESB mit 2 ungekoppelt Leitungselementen und 2 idealen Transformatoren

Leerlaufendes Unity Element is kapazitiv KS Unity

Element is induktiv

